

Week 5 work

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1 Confirmation of ML Problem definition

Prediction problem: to estimate the probability that a patient will fail to attend a scheduled appointment ("No-Show") at the time the booking is created

Proposed Target: a binary variable where 1 represents a "No-Show" and 0 represents a successful attended outcome

Candidate Features:

- Demographics: Gender and Age group
- Appointment Details: Type, day of the week, time and distance to the clinic
- Behavioral History: lead days (time between booking and appointment), total previous appointments and historical no-show count
- Intervention Data: Reminder status and channel

Treatment of Cancellations: Currently "Cancelled" appointments are grouped with attended appointments in the negative class

Modelling Objective: to reduce clinic revenue loss and maximize resource utilization by identifying high-risk appointments to trigger automated secondary reminders

2 Week 5 Analysis: Recommended Changes and Justifications

Upon further analysis of the data and the business objective I recommend the following change to the problem definition

2.1 Treatment of Cancellations

Current Decision: Group "Cancelled" with "Attended" (Class 0)

Proposed change: exclude "Cancelled" appointments from the training and evaluation datasets entirely. The model should strictly compare No-Show (1) versus Attended (0)

Justification:

Business Logic:a No-Show is a total loss of a time slot.A Cancellation is a proactive patient action that allows the clinic a chance to re-book the slot.Mixing these two behaviors in Class 0 creates a noisy baseline

Predictive Accuracy:the behavioral drivers for someone who calls to cancel(responsible,schedule-aware) are fundamentally different from someone who simply fails to show up (forget,lack of transport).Training the model to treat them the same will likely decrease the model's ability to identify true no-show risks

2.2 Feature Engineering(Intervention Data)

Current Decision:Use reminder_sent and reminder_channel as candidate features

Proposed Change:Remove these as input features for the initial prediction made at the time of booking

Justification:

Data Leakage/Timing:the document states the model predicts at the time of booking.At that specific moment,a reminder has not yet been sent.Including this as a feature assumes knowledge of the future

Objective Alignment:the goal is to use the model's output to decide whether to send a reminder.If the reminder is an input you create a circular dependency that will make the model's performance look better than it actually is in a production environment.

3 Data Preparation

- Strict Target:I filtered out "Cancelled" appointments.By training on a clean dataset of only people who actually had the opportunity to show up,the model becomes much more precise
- Leakage Prevention:dropping waiting_time_minutes.In many hospitals datasets this is the "false" predictor because it has 100% correlation with attending
- Time Features:converting appointment_time into a numeric score because there is a logical progression from morning to evening that the model can learn from.
- Imputation:using the median for distance.The median is safer than the mean in healthcare data because clinic distances often have outliers(patients travelling from very far away)

4 Data Exploration and Visual Evidence

4.1 Target Class Distribution

This shows the ratio of patients who showed up versus those who did not. We observe that there is a class imbalance hence the decision will be that accuracy will not only be the metric to be used and also use recall for the No-Show class (to ensure we catch as many potential no-shows as possible) and use the F1-Score for evaluation.

4.2 Booking Lead Days Box Plot

This shows the spread of days waited for both groups. We observe that the No-Show group (Target 1) has a significantly higher median and upper quartile lead time compared to the attended group hence this justifies `booking_lead_days` as a top-tier feature for selection. It suggests that the longer a patient has to wait for their appointment the more likely they are to forget or have a schedule conflict.

4.3 Distance to Clinic Violin Plot

It shows the density of patient distances for both groups. We observe that the distribution is relatively similar for both classes though no-shows have slightly thicker tails at longer distances hence we conclude that distance is a weak but useful feature since there are no extreme outliers affecting one class specifically hence we will keep it as a numeric feature without further transformation.

4.4 Correlation Heatmap

This shows the linear relationship between all numeric inputs and the target. We observe that there are no features showing a correlation of greater than 0.8 with one another meaning no redundant features.

This validates our data leakage removal and the feature selection is now good and the model will not rely on future information.

5 Feature Engineering

Three new features were created:

- `no_show_rate`: a ratio was calculated as `previous_no_shows` divide by `previous_appointments`
A patient with 2 no-shows out of 2 visits is higher risk than a patient with 2 no-shows out of 50 visits. Expected modelling value is expected to be very high.
- `is_weekend`: a binary flag (1 for saturday/sunday, 0 for weekdays) derived from `appointment_day`.

Patient logistics such as work,school.public transport availability change significantly on weekends and the expected modelling value is expected to be medium as it captures scheduling patterns that differ from the monday to friday work week.

- is_urgent :a binary flag(1 if booking_lead_days is 0, otherwise 0).Patients who book an appointment for the same day they visit usually have an immediate medical need.The expected modelling value is high as it creates a low risk indicator as urgent needs drastically reduce the chance of a forgetful no-show

6 Train/Test Strategy

Strategy:Stratified Random Split(80/20)

I split the dataset into 80% for training the model and 20% for testing its performance on unseen data.We used stratification on the target variable.

The strategy is appropriate because of Class Imbalance:Earlier observation showed that No-Shows are less frequent than attended outcomes.Stratification ensures that the training set and the test set both contain the same percentage of No-Shows as the original data.

An 80/20 split provides enough data for the model to learn patterns while reserving a significant portion to validate measure how well the model works in the real world.

7 Baseline Model

Selected Model:Logistic Regression

The selected model is simple,highly interpretable and computationally efficient.It establishes the floor for performance hence any complex model I build later should perform better than this

I scaled the numeric features which are essential for logistic regression and fit the model to the training data and then evaluate its ability to identify the "No-Show" class on the test set.

7.1 Evaluation strategy

Accuracy:used to see overall correctness but I expect this to be high because most people attend their appointments.The model had a score of 0.62 hence the model correctly predicts the outcome for 62% of all appointments in the test set.

Recall:this tells us of the actual No-Shows what percentage did the model correctly flag?With a score of 0.67 this is a primary success metric and the model successfully identifies 67% of all actual No-Shows

Precision:this tells us that when the model predicts a No-Show how often is it actually correct and is important to ensure we are not annoying patients with

unnecessary reminders. With a score of 0.62 we conclude that when the model flags an appointment as a No-Show risk it is correct 62% of the time.

F1-Score: this is the balance between precision and recall

ROC-AUC: measures the model's ability to distinguish between classes. 0.6682 was the score and it indicates that there is a 67% probability that the model will rank a random No-Show higher than a random attended patient. This is a solid foundation for a baseline.

7.2 Class Analysis

No-Show Class (Target): The model prioritizes Recall (0.67) over Precision (0.62). From a clinical operations perspective, this is a favorable trade-off: we would rather send a few extra reminders to people who might have attended (False Positives) than fail to identify people who will actually miss their slot (False Negatives).

Attended Class: The model correctly identifies 56% of patients who will attend. The balanced F1-scores (0.59 vs 0.64) indicate that the `class_weight='balanced'` parameter successfully prevented the model from simply defaulting to the majority class.

7.3 Business Impact and Observation

Efficiency: By using this model, the clinic can proactively target interventions like secondary SMS or phone calls at the 67% of high-risk patients identified by the model.

Room for Growth: An ROC-AUC of 0.67 suggests that while the linear relationships (like lead days and previous history) are being captured, there are likely non-linear complexities (like interactions between age and appointment type) that Logistic Regression cannot fully grasp.

8 Recommendations

Model Complexity: transition to non-linear tree based models like random forest to see if they can improve the ROC-AUC score to greater than 0.75

Feature Selection: Re-evaluate features with low correlation like `is_weekend` to see if removing them reduces noise and improves precision

Hyperparameter Tuning: optimizing the decision threshold to further increase Recall if the clinic has the capacity to handle more outgoing reminders.